# Experimentation Results

# What I did

- Pipeline: $C_1$ finds objects (YOLOv11n), $C_2$ predicts their paths (AutoBot-Ego style), $C_3$ plans the ego path (PlanT style). All three are ML models.
- Data: 350 nuScenes scenes, front camera. $C_1$ is trained in Boston, $C_2$ and $C_3$ on other Boston scenes, $C_1$ is updated on Singapore, and all tests use the same 50 Singapore scenes (1,720 test samples).
- Two old $C_1$ (seed 0 and seed 1). Each is updated 15 times (100/50/25 % of the data, 5/10/20 epochs, 3 seeds): **30 updates**.
- $C_2$ and $C_3$ are trained once in two ways and never retrained during the updates: **old-pipeline training** (learn from the old $C_1$ output, including its mistakes) and **isolated training** (learn from the ground truth).
- For each update: $\delta_1$ = gain of $C_1$ (mAP50), $\Delta_2$ = increase of the prediction error, $\Delta_3$ = increase of the planning error at 3 s (metres).

# Result 1: every update made $C_1$ better

All 30 updates increased mAP50 ($\delta_1$ from +0.048 to +0.118; old $C_1$ 0.109 and 0.117, updated $C_1$ 0.166 to 0.235).

# Result 2: $C_2$ and $C_3$ work (checked against simple baselines)

| Model | Error on the test scenes | Simple baseline |
|---|---|---|
| $C_2$ (input: ground truth) | 0.77 m | 4.48 m (objects stay still) |
| $C_2$ (input: $C_1$ output) | 2.39 to 2.95 m | 5.39 to 5.85 m |
| $C_3$ without the ego path | 9.3 to 12.5 m | 9.25 m (average path) → dropped |
| $C_3$ with the last 1.5 s ego path | 3.70 to 4.16 m, collisions 5–8 % | 9.25 m, collisions 18 % |

$C_3$ still uses the objects: removing them moves its plan by 0.73 to 1.65 m.

# Result 3 (main): a better $C_1$ harms the $C_2$ that learned its old mistakes

Number of updates (out of 30) in which the error increased. "Above noise" = larger than the change caused by only swapping the old $C_1$ seed (0.112 m prediction, 0.018 m planning). "Significant" = 95 % bootstrap interval over the 50 test scenes above zero.

| | Prediction ($C_2$) | | Planning ($C_3$) | |
|---|---|---|---|---|
| | old-pipeline | isolated | old-pipeline | isolated |
| Any increase | **23** | 5 | 18 | 29 |
| Above noise | **11** | 0 | 13 | 29 |
| Significant | **3** | 0 | 1 | 10 |

- In **30 of 30** updates, the old-pipeline $C_2$ got worse than the isolated $C_2$. Same input, different training: the harm comes from learning the old mistakes.
- All 11 updates above noise start from the old $C_1$ with seed 0 (mean $\Delta_2$ = +0.165 m). The $C_2$ of seed 0 is also hurt by a simple seed swap (+0.112 m) although both old $C_1$ have almost the same accuracy (0.009 mAP50 apart): it reacts to the mistakes, not to the accuracy.
- Planning changes very little (mean $\Delta_3$ −0.006 to +0.058 m, at most 0.13 m of about 4 m), and more often for isolated training: this is not caused by learned mistakes.
- By the simple rule ($\delta_1 > 0$ and $\Delta_2 > 0$ or $\Delta_3 > 0$), entangled enhancement appears in **29 of 30** updates.

## Result 4: retraining $C_2$ and $C_3$ on the new $C_1$ lowers both errors

Tested on 2 updates (new-pipeline training). Prediction error 2.75 / 2.61 m → 2.39 / 2.45 m, planning error 4.14 / 4.00 m → 3.71 / 3.70 m. Both are even below the errors before the update.

## Result 5: no warning sign found yet (H2 not supported)

Nine measures of how much the output of $C_1$ and $C_2$ changes (no ground truth needed) were tested. The best one, the shift of $C_2$'s predictions, looks good over all 30 updates (rank correlation +0.68, AUROC 0.81), but within each old $C_1$ it is +0.29 and −0.28: it only separates the two starting models. The accuracy gain $\delta_1$ does not predict the harm either (rank correlation +0.04). Hint for the next step: the old $C_1$ whose $C_2$ reacted to the seed swap is the one whose updates caused harm.

## All 30 updates

Old-pipeline (op) and isolated (iso) training, $C_3$ with the ego path. Shaded = above noise. Last column: significant harm (bootstrap) for op.

| # | Update (base, data, epochs, seed) | $\delta_1$ | $\Delta_2$ op | $\Delta_2$ iso | $\Delta_3$ op | $\Delta_3$ iso | Sig. |
|---|---|---|---|---|---|---|---|
| 1 | s0, 100%, 10 ep, seed 1 | +0.077 | +0.049 | −0.141 | −0.027 | +0.035 | – |
| 2 | s0, 100%, 10 ep, seed 2 | +0.096 | +0.062 | −0.212 | +0.026 | +0.067 | – |
| 3 | s0, 100%, 10 ep, seed 3 | +0.072 | +0.139 | −0.060 | +0.020 | +0.078 | – |
| 4 | s0, 50%, 10 ep, seed 1 | +0.073 | +0.184 | +0.021 | −0.012 | +0.081 | – |
| 5 | s0, 50%, 10 ep, seed 2 | +0.077 | +0.192 | +0.100 | +0.031 | +0.063 | – |
| 6 | s0, 50%, 10 ep, seed 3 | +0.065 | +0.098 | −0.154 | +0.040 | +0.127 | – |
| 7 | s0, 25%, 10 ep, seed 1 | +0.064 | +0.233 | −0.058 | −0.030 | +0.020 | – |
| 8 | s0, 25%, 10 ep, seed 2 | +0.059 | +0.284 | +0.082 | −0.042 | +0.071 | $\Delta_2$ |
| 9 | s0, 25%, 10 ep, seed 3 | +0.067 | +0.157 | −0.141 | −0.046 | +0.034 | – |
| 10 | s0, 100%, 5 ep, seed 1 | +0.094 | +0.182 | −0.073 | −0.024 | +0.052 | – |
| 11 | s0, 100%, 5 ep, seed 2 | +0.101 | +0.284 | +0.001 | −0.006 | +0.040 | $\Delta_2$ |
| 12 | s0, 100%, 5 ep, seed 3 | +0.109 | +0.266 | +0.093 | −0.042 | +0.033 | $\Delta_2$ |
| 13 | s0, 100%, 20 ep, seed 1 | +0.069 | +0.107 | −0.079 | −0.000 | +0.051 | – |
| 14 | s0, 100%, 20 ep, seed 2 | +0.080 | +0.127 | −0.051 | +0.025 | +0.059 | – |
| 15 | s0, 100%, 20 ep, seed 3 | +0.103 | +0.115 | −0.085 | +0.001 | +0.056 | – |
| 16 | s1, 100%, 10 ep, seed 1 | +0.068 | −0.054 | −0.170 | −0.001 | +0.021 | – |
| 17 | s1, 100%, 10 ep, seed 2 | +0.066 | −0.081 | −0.156 | +0.026 | +0.073 | – |
| 18 | s1, 100%, 10 ep, seed 3 | +0.065 | −0.187 | −0.280 | +0.023 | +0.026 | – |
| 19 | s1, 50%, 10 ep, seed 1 | +0.065 | −0.030 | −0.115 | +0.072 | +0.084 | $\Delta_3$ |
| 20 | s1, 50%, 10 ep, seed 2 | +0.067 | −0.135 | −0.152 | +0.054 | +0.088 | – |
| 21 | s1, 50%, 10 ep, seed 3 | +0.079 | −0.089 | −0.141 | +0.037 | +0.070 | – |
| 22 | s1, 25%, 10 ep, seed 1 | +0.048 | +0.090 | −0.059 | −0.001 | +0.029 | – |
| 23 | s1, 25%, 10 ep, seed 2 | +0.078 | +0.081 | −0.095 | −0.042 | −0.004 | – |
| 24 | s1, 25%, 10 ep, seed 3 | +0.063 | +0.082 | −0.111 | +0.010 | +0.022 | – |
| 25 | s1, 100%, 5 ep, seed 1 | +0.086 | −0.025 | −0.267 | +0.033 | +0.068 | – |
| 26 | s1, 100%, 5 ep, seed 2 | +0.118 | +0.040 | −0.094 | +0.003 | +0.064 | – |
| 27 | s1, 100%, 5 ep, seed 3 | +0.115 | +0.052 | −0.107 | +0.015 | +0.050 | – |
| 28 | s1, 100%, 20 ep, seed 1 | +0.102 | +0.003 | −0.094 | +0.049 | +0.047 | – |
| 29 | s1, 100%, 20 ep, seed 2 | +0.071 | +0.065 | −0.029 | +0.021 | +0.022 | – |
| 30 | s1, 100%, 20 ep, seed 3 | +0.071 | +0.073 | −0.025 | +0.018 | +0.049 | – |

## Bottom line

- Entangled enhancement is real for the prediction module: a better $C_1$ harms the $C_2$ that learned the old $C_1$, and not the $C_2$ trained on the ground truth.
- The effect on planning is very small.
- Retraining $C_2$ and $C_3$ together with $C_1$ lowers both errors, but no cheap warning sign has been found yet. Early detection is the next step.